

Hacks and tweaks

You could adapt this game in many ways. You can show the number of moves taken to finish the game, so the player can try and beat their own score or challenge their friends. You could also add more symbols to make the game harder.

Show the number of moves

At the moment, the player has no way of knowing how well they've done or if they've done any better than their friends. How can we make the game more competitive? Let's add a variable to count how many turns a player takes to finish the game. Then players can compete to see who gets the lowest score.

1 Add a new module

You need to import Tkinter's `messagebox` widget to display the number of moves at the end of the game. In the import line, add the word `messagebox` after `DISABLED`.

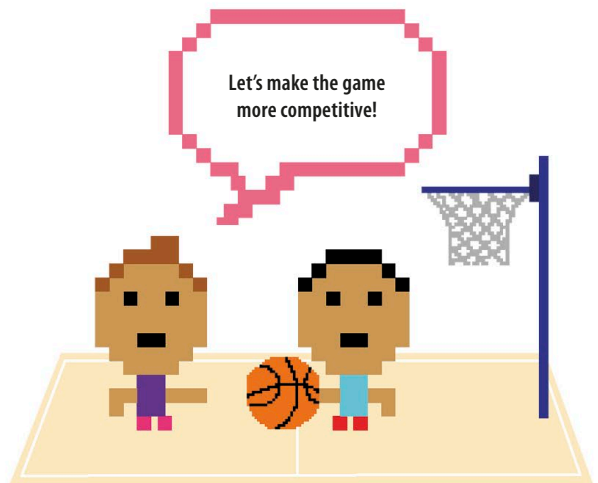
```
from tkinter import Tk, Button, DISABLED, messagebox
```

2 Make new variables

You'll have to make two extra variables for this hack. One variable will keep track of the number of moves the player makes, while the other will remember how many pairs they've found. Give them both a starting value of 0. Put these lines below the variable `previousY`.

```
previousY = 0
moves = 0
pairs = 0
```

The player hasn't made any moves yet, or found any pairs, so the values are 0.



3 Declare them global

The `moves` and `pairs` variables are global variables, and they'll need to be changed by the `show_symbol()` function. Let `show_symbol()` know this by putting these two lines near the top of the function.

```
def show_symbol(x, y):
    global first
    global previousX, previousY
    global moves
    global pairs
```